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Large-Scale Data Sources for Redox Mediators and Organic Anolytes in Flow Batteries

Executive summary

The strongest **machine-readable starting points** for this topic are, in order, **D3TaLES**, **RedDB**, and **SOMAS**. D3TaLES is the broadest organic redox-molecule infrastructure here, with a 2025 final public export of **35,729 molecules** and an article describing **more than 43,000** curated redox-active organic molecules; its schema explicitly includes oxidation/reduction potentials, solubility, stability, diffusion, conductivity, and literature links. RedDB is the most useful **aqueous flow-battery-specific computed screening database**, with **31,618 quinones and aza-aromatics**, downloadable in relational/CSV/XLSX form. SOMAS is the best large **experimental aqueous-solubility** database in this space, with **11,696 organic compounds**, but it is solubility-centric rather than a full electrochemistry/lifetime resource. [1]

The most valuable **experimentally complete compound-level table** I found is still the older but unusually rigorous **Wedege et al.** paper in *Scientific Reports* (2016), which experimentally studies **33 compounds** and provides supplementary tables with **formal potentials at multiple pH values, solubility in multiple electrolytes, and qualitative stability/electrochemical reversibility comments**. Newer reviews are more current on failure modes and design rules, but most do **not** expose equivalently clean machine-readable compound tables with standardized stability fields. [2]

For **stability interpretation**, the most important review remains **Kwabi, Ji, and Aziz, *Chemical Reviews* (2020)**, which organizes decomposition mechanisms across five major classes. For **recent chemistry-specific stability context**, the best follow-on reviews are the **2024 viologen review**, the **2025 quinone degradation review**, and the **2024 AORFB perspective**. For **frontier screening**, **RedCat (2025)** is notable because it screened **112 million PubChem molecules** and shortlisted **261 promising anolyte candidates**, though I could not verify a public release of the full shortlisted candidate table during this pass. [3]

The biggest practical obstacle is that **"stability" is not reported consistently** across sources. Depending on the paper, it may mean qualitative reversibility; capacity retention after *N* cycles; fade in % per day, % per cycle, or % per hour; symmetric-cell vs full-cell results; or individual decomposition pathways/half-lives. Solubility units also vary between **M, mg/L, and logS**, often without consistent pH and ionic-strength normalization. That heterogeneity makes a single cross-source "potential vs stability" scatter plot more likely to mislead than to clarify unless one first normalizes the raw data. [4]

Scope and ranking logic

I prioritized sources that are strongest on the fields you asked for: **compound identity, redox potential, stability metrics, and solubility/concentration limits**, with extra weight for **public supplementary tables, downloadable datasets, APIs, or relational exports**. In practice, the field splits into three layers: large machine-readable databases; review papers that synthesize degradation and design rules; and primary papers or patents that contain rich examples but often require manual extraction from SI. Dedicated large datasets specifically labeled **redox mediators** are much scarcer than broader datasets of **organic electroactive molecules** or **organic anolytes**; that is an inference from the fact that the major open databases and most recent reviews are organized around electroactive molecules/anolytes rather than mediator-only taxonomies, while the main redox-mediated battery review is conceptually useful but not data-table-centric. [5]

Ranked source inventory

Highest-value academic and database sources

Rank	Source	Year	Type	Scale	What it gives you	Best access point
Top	**D3TaLES** — Duke, Bhat, Sornberger, Odom, Risko, et al., *Digital Discovery*	2023	Database + API + bulk export	**35,729 public molecules** in final 2025 export; article describes **>43,000 curated molecules**	Molecule-centric schema with oxidation/reduction potential, solubility, stability, conductivity, diffusion coefficient, charge-transfer rate, literature links, and raw/processed separation; strongest general-purpose infrastructure here	Final public dataset, API, and documentation all on the D3TaLES site. [6]

Rank	Source	Year	Type	Scale	What it gives you	Best access point
Very high	**RedDB** — Sorkun, Zhang, Khetan, Sorkun, Er, *Scientific Data*	2022	Relational computational database	**31,618 molecules** from **52 core molecules**; **15 tables**	Quinones + aza-aromatics; structural, thermodynamic, electronic, and reaction data; ML-predicted aqueous solubility; downloadable reduced forms in CSV/XLSX from Harvard Dataverse; ideal for high-throughput filtering	Article + Harvard Dataverse record + GitHub schema/repo. [7]
High	**SOMAS** — Gao, Andersen, Sepulveda, Panapitiya, Murugesan, Wang, et al., *Scientific Data*	2022	Experimental solubility database	**11,696 compounds**; **26 CSV columns**	Experimental aqueous solubility plus temperature, references, five identifiers, eight DFT descriptors, six molecular descriptors, and XYZ geometries; excellent for concentration-limit modeling	Article + official Figshare item containing dataset and XYZ files. [8]

Rank	Source	Year	Type	Scale	What it gives you	Best access point
High	**Organic Redox Species in Aqueous Flow Batteries: Redox Potentials, Chemical Stability and Solubility** — Wedege, Dražević, Konya, Bentien, *Scientific Reports*	2016	Primary experimental paper + rich SI	**33 compounds**	Probably the single best compound-by-compound experimental table for potentials, solubility in multiple electrolytes, and qualitative stability/ reversibility comments; includes selected flow-cell outcomes	Main article + Springer supplementary PDF tables S4–S5. [2]
High	**RedCat, an automated discovery workflow for aqueous organic electrolytes** — Sorkun, Zhou, Murigneux, Menegazzo, Er, et al., *Digital Discovery*	2025	Screening workflow paper	**112 million PubChem molecules screened**; **261 promising anolyte candidates** shortlisted	ML redox reaction energy, ML aqueous solubility, pricing integration, first-principles validation, experimental testing of two molecules; excellent for current screening strategy	Open-access article; ESI is indicated on the publisher page, but a public full candidate table was not confirmed in this pass. [9]

Rank	Source	Year	Type	Scale	What it gives you	Best access point
High	**Electrolyte Lifetime in Aqueous Organic Redox Flow Batteries: A Critical Review** — Kwabi, Ji, Aziz, *Chemical Reviews*	2020	Review	Class-level, not enumerated in a machine-readable table	Best synthesis of decomposition pathways and lifetime interpretation; spans **quinones, viologens, aza-aromatics, iron coordination complexes, and nitroxide radicals**	ACS landing page and NSF-hosted public-access manuscript. [10]
Medium-high	**Family Tree for Aqueous Organic Redox Couples for Redox Flow Battery Electrolytes** — Fischer, Mazúr, Krakowiak, *Molecules*	2022	Conceptual review	Class-level map rather than explicit dataset	Good entry point for chemical-family coverage and literature navigation; useful to discover less common redox families	MDPI article / PMC mirror. [11]
Medium-high	**Viologen-based aqueous organic redox flow batteries: materials synthesis, properties, and cell performance** — Yin, Duanmu, Liu, *J. Mater. Chem. A*	2024	Chemistry-specific review	Not explicitly enumerated	Best focused review for viologen negolytes; useful for solubility/ stability structure–property tradeoffs in a major anolyte family	RSC article landing page. [12]

Rank	Source	Year	Type	Scale	What it gives you	Best access point
Medium-high	**Quinones: understanding their electrochemistry, chemistry and degradation pathways to tap their full potential in aqueous redox flow batteries** — Afzal, Liyanaarachchi, Collis, Jones, Maniam, *J. Mater. Chem. A*	2025	Chemistry-specific review	Not explicitly enumerated	Most current quinone-focused degradation review I found; particularly useful for failure pathways and substituent effects	RSC article page. [13]
Medium	**Perspectives on aqueous organic redox flow batteries** — Zhu, Chen, Fu, *Green Energy & Environment*	2024	Perspective/ review	Not explicitly enumerated	Strong recent overview of design principles, degradation mechanisms, characterization, and computational/ ML directions	Open-access article page. [14]

Additional sources worth checking immediately

A useful near-follow-up source is **"Computational design of quinone electrolytes for redox flow batteries using high-throughput machine learning and theoretical calculations"**. The paper reports screening **100,000 di-substituted quinones**, while the visible Figshare "DataSheet1" appears to expose **1,517 small quinones**. That mismatch makes it valuable but also something you should verify before using it as a canonical raw dataset. [15]

Relevant patent families worth mining

Patent family	Why it matters for data extraction	Caveat
US11056705B2 — Organic anolyte materials for flow batteries	Broad organic-anolyte patent family; useful for enumerating claimed structures and examples	Patent formatting is poor for machine extraction and often mixes claim breadth with sparse standardized measurements. [16]
US11557786B2 / WO2020072406A2 — Extending the lifetime of organic flow batteries via redox state management	Important for anthraquinone lifetime management and capacity-loss mitigation strategies	More operational than descriptor-rich; useful for stability interventions rather than a clean compound table. [17]
US20180072669A1 / US10934258B2 — Materials for use in an aqueous organic redox flow battery	Viologen- and metallocene-based materials; useful for structure lists and early viologen family coverage	Still legal-document oriented; not a substitute for a curated dataset. [18]

Representative compound entries

The table below is intentionally **condition-specific**. Most electrochemical values are **formal potentials vs NHE** reported under particular pH/electrolyte conditions in the source tables, so they should not be mixed without normalization. I emphasize compounds for which the source also includes either qualitative stability comments or explicit battery-cycling observations. [19]

Compound	Chemistry	Relevant potential from source	Solubility / concentration limit from source	Stability metric reported in source	Notes
AQDS(2,7) [20]	Sulfonated anthraquinone	**E°′ = −0.281 V** at pH 13; **0.165 V** at pH 0	**0.49–0.61 M** in 1 M KOH; **0.81–1.21 M** in 1 M H_2SO_4	Quasireversible at pH 0 and 7; reversible at pH 13; comment: **thixotropic behaviour**	One of the strongest benchmark aqueous anthraquinones.
AQDS(1,5) [20]	Sulfonated anthraquinone	**E°′ = −0.501 V** at pH 13	**<0.01 M** in 1 M KOH; **0.11–0.12 M** in 1 M H_2SO_4	Quasireversible at pH 0 and 7; reversible at pH 13; **phase separation upon standing**	Low-potential but severely solubility-limited in alkaline media.

Compound	Chemistry	Relevant potential from source	Solubility / concentration limit from source	Stability metric reported in source	Notes
AQS(2) [21]	Monosulfonated anthraquinone	**E°′ = −0.334 V** at pH 13; **0.163 V** at pH 0	**0.29–0.36 M** in 1 M KOH; **0.54–0.64 M** in 1 M H_2SO_4	Reported as **reversible**; comment: **phase separation upon standing**	Good counterexample showing why higher nominal solubility is not enough if phase behavior is poor.
2,6-DHAQ / AQDH(2,6) [22]	Hydroxylated anthraquinone	**E°′ = −0.701 V** at pH 13	**0.52–0.59 M** in 1 M KOH; **<0.01 M** in acid	**Reversible at pH 13**	Classic strong alkaline anolyte benchmark.
1,5-DHAQ / AQDH(1,5) [22]	Hydroxylated anthraquinone	**E°′ = −0.520 V** at pH 13	**0.13–0.15 M** in 1 M KOH	**Quasireversible at pH 13**; electrode reaction / multiple-peak issues noted	Lower-performing isomer relative to 2,6-DHAQ.
Rhein / AQDH(4,5)CA [23]	Carboxylated anthraquinone	**E°′ = −0.546 V** at pH 13	**0.35–0.39 M** in 1 M KOH	Quasireversible at pH 7; reversible at pH 13; multiple peaks and thixotropic behaviour noted	Attractive because of good alkaline solubility and low potential, but not perfectly clean electrochemistry.
NQ(1,2)S [24]	Sulfonated naphthoquinone	**E°′ = −0.081 V** at pH 13; **0.55 V** at pH 0	**0.77–0.96 M** in 1 M KOH	Quasireversible in SI; in a flow cell with ferrocyanide, the source reports **significant capacity loss over 90 cycles**	Good illustration that high alkaline solubility does not guarantee cycling stability.

Compound	Chemistry	Relevant potential from source	Solubility / concentration limit from source	Stability metric reported in source	Notes
Lawsone / NQ(1,4)H [24]	Hydroxynaphthoquinone	**E°′ = −0.48 V** at pH 13	**0.82–0.96 M** in 1 M KOH	Quasireversible in SI; in a flow cell with ferrocyanide, the source reports **significant capacity loss after only two cycles**	Important cautionary example of apparently promising descriptors masking poor practical stability.
Hydroquinone / HQ(1,4) [25]	Benzoquinone/ hydroquinone	**E°′ = 0.669 V** at pH 0; **0.433 V** at pH 7	**1.5–1.8 M** in 1 M KOH; **0.29–0.34 M** in 1 M H_2SO_4	Quasireversible at pH 0 and 7; pH 13 value estimated from first reduction peak	Strong solubility, but the benzoquinone family is generally more stability-limited.
Phenazine [26]	Aza-aromatic	**E°′ = 0.403 V** at pH 0	**0.01 M** in 1 M H_2SO_4; **<0.01 M** in 1 M KCl	**Unstable at pH 7** in this source	Useful as a family marker, but the parent molecule is far less practical than modern substituted phenazines.
TEMPOL [27]	Nitroxide radical	**E°′ = 0.803 V** at pH 7; **0.73 V** at pH 0	**1.5 M** in 1 M KCl; **2.5 M** in 1 M H_2SO_4	Source reports **no degradation after 30 days in aqueous buffer in the uncharged state**, but also **significant capacity loss after a limited number of battery cycles**	Excellent example of the difference between chemical shelf stability and full-cell cycling stability.

The clearest pattern in these examples is that **alkaline anthraquinones** still dominate the “low potential + useful aqueous solubility” region, while **naphthoquinones and benzoquinones** often offer either higher potential or good solubility but poorer practical stability. Nitroxide radicals such as TEMPOL can be highly soluble and chemically persistent in some states, yet still underperform in flow-cell cycling. [28]

Dataset quality issues and gaps

The biggest structural gap is that **no single public resource simultaneously covers large compound counts and standardized experimental lifetime data**. D3TaLES is the richest infrastructure and explicitly supports stability/solubility/electrochemistry at the schema level, but its public bulk exports are described in terms of completed **calculations** for key properties, so experimental stability coverage appears uneven relative to computed fields. RedDB is large and well organized but its aqueous solubility is **ML-predicted** rather than experimental. SOMAS solves the solubility problem, but it does **not** provide measured cycling stability or redox potentials as primary fields. [29]

There are also several important comparability issues. Wedege et al. report **formal potentials vs NHE** across multiple pH values and electrolytes, plus qualitative reversibility/stability notes, which is experimentally valuable but hard to merge directly with databases that use computed reaction energies, predicted logS, or non-standardized literature extraction. Recent reviews make this explicit by centering the field around the triad of **redox potential, solubility, and stability**, yet they still mostly summarize results in prose, figures, and family-level discussions rather than a normalized machine-readable matrix. [30]

A more specific quality issue is that **public workflow outputs are not always congruent with paper claims**. The 2022 Frontiers quinone-design paper reports screening **100,000 di-substituted quinones**, but the visible Figshare “DataSheet1” excerpt points to **1,517 small quinones**. That does not invalidate the work, but it means you should verify whether the public file is a filtered subset, a training set, or the actual screening output before using it as a raw descriptor table. [15]

Finally, dedicated large datasets specifically for **soluble redox mediators in redox-targeting / redox-mediated flow batteries** remain sparse. The main broad review of aqueous organic and redox-mediated flow batteries is conceptually helpful for mediator selection, but it is not a mediator-by-mediator open database comparable to D3TaLES or RedDB. [31]

Publication timeline and data-acquisition next steps

The timeline below captures the most important review/database milestones for this specific data-collection problem. It emphasizes when the field moved from small experimental tables to open-data infrastructures and then to very large automated screening workflows. [32]

```
timeline
    title Key review and database milestones for organic flow-battery molecule
data
    2016 : Wedege et al. publish an experimental 33-compound table with SI for
potentials, solubility, and qualitative stability
    2020 : Kwabi, Ji, Aziz publish the Chemical Reviews critical review on
electrolyte lifetime and decomposition
    2022 : Family Tree review maps aqueous organic redox-couple families
         : RedDB and SOMAS publish open data descriptors for aqueous flow-
battery discovery
    2023 : D3TaLES publishes its redox-organic data infrastructure
    2024 : Perspectives on AORFBs updates degradation and characterization
```

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directions
         : Viologen-focused review consolidates negolyte design rules
     2025 : Quinone-focused degradation review deepens stability interpretation
         : RedCat screens 112 million PubChem molecules and shortlists 261
anolyte candidates
```

If the goal is to build a **usable raw dataset** now, the most defensible next sequence is this: start from **D3TaLES + RedDB + SOMAS** as the backbone; add **Wedege 2016** as the highest-value experimental table for pH-dependent potentials/solubility/stability annotations; then layer in **manual extraction of stability metrics** from the Chem Rev lifetime review, the viologen review, the quinone degradation review, and selected recent primary papers. That hybrid route will be far cleaner than trying to scrape all recent reviews first. [33]

If raw data are missing or incomplete, the most practical next steps are:

- use the **D3TaLES REST API** and final public bulk export to pull all available molecule-centric records, including literature links and any experimental stability fields exposed in the schema; the documentation also gives a direct contact route for questions or contributions; [34]
- download **RedDB** from the Harvard Dataverse and preserve the relational structure, because the reaction-table context matters for aqueous quinone/aza-aromatic comparisons; [35]
- download **SOMAS** from Figshare, then normalize its units alongside RedDB/D3TaLES so that solubility is stored in both native and converted forms; [36]
- for reviews and patents that remain non-machine-readable, request the authors' **spreadsheet source tables** or parse SI PDFs only after fixing a common schema for **reference electrode, pH, solubility unit, cell format, and stability metric definition**. [37]

The main limitation of this report is that a few high-value recent workflow papers clearly exist but their **full public candidate tables were not verifiably exposed** from the accessible landing pages during this pass. The most notable examples are **RedCat** and the **Frontiers quinone-design workflow**. I have therefore treated them as **important but partially incomplete data sources**, rather than overstating their immediate downloadability. [38]

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[1] [6] [33] https://pubs.rsc.org/en/content/articlelanding/2023/dd/d3dd00081h
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